

```
import pandas as pd

import matplotlib.pyplot as plt


data = {
    'Customer_ID': [101, 102, 103, 104, 105, 106, 107, 108],
    'Age': [22, 35, 26, 45, 52, 23, 40, 30],
    'Spending': [2500, 8000, 1500, 12000, 3000, 500, 9500, 4000],
    'Visits': [15, 40, 10, 55, 18, 5, 48, 20]
}


df = pd.DataFrame(data)


print("Original Dataset:\n")
print(df)


def segment_customer(spending):
    if spending >= 8000:
        return 'High'
    elif spending >= 3000:
        return 'Medium'
    else:
        return 'Low'


df['Segment'] = df['Spending'].apply(segment_customer)


print("\nCustomer Segments:\n")
print(df)
```

```
high_value = df[df['Segment'] == 'High']
```

```
print("\nHigh Value Customers:\n")
```

```
print(high_value)
```

```
low_engagement = df[df['Visits'] < 15]
```

```
print("\nLow Engagement Customers:\n")
```

```
print(low_engagement)
```

```
plt.figure(figsize=(8,5))
```

```
plt.hist(df['Spending'], bins=5)
```

```
plt.title("Spending Distribution")
```

```
plt.xlabel("Spending Amount")
```

```
plt.ylabel("Number of Customers")
```

```
plt.show()
```

```
segment_count = df['Segment'].value_counts()
```

```
plt.figure(figsize=(6,6))
```

```
plt.pie(segment_count,
```

```
    labels=segment_count.index,
```

```
    autopct='%1.1f%%')
```

```
plt.title("Customer Categories")
```

```
plt.show()
```

```
print("\nBusiness Strategies:")
```

```
print("1. Offer premium rewards to High-value customers.")
```

```
print("2. Send discounts and offers to Low-engagement users.")
```

```
print("3. Create personalized marketing campaigns.")
```

```
print("4. Increase engagement through loyalty programs.")
```

